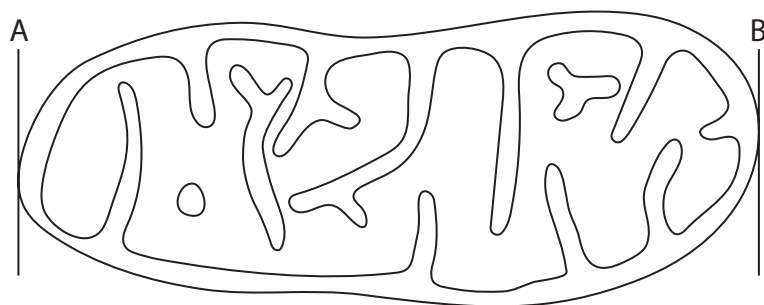


7 In eukaryotic organisms aerobic respiration takes place in mitochondria.

(a) The diagram shows a mitochondrion.



The actual length of the mitochondrion between A and B was $0.75\text{ }\mu\text{m}$.

Calculate the magnification of the mitochondrion shown in the diagram.

Give your answer in standard form.

(2)

Answer

(b) In an investigation, the rate of production of carbon dioxide by mitochondria was measured.

Three samples of mitochondria had the following chemicals added to them:

- Sample 1: pyruvate
- Sample 2: pyruvate and ADP
- Sample 3: pyruvate and pyruvate transport inhibitor.

The table shows the results of this investigation.

Sample	Chemicals added	Rate of carbon dioxide production in 20 minutes / nmol per mg of mitochondria
1	pyruvate	62.6
2	pyruvate and ADP	85.0
3	pyruvate and pyruvate transport inhibitor	11.6

(i) Which is the approximate rate at which carbon dioxide is produced in sample 1?

(1)

- ☐ **A** 3 nmol mg min
- ☐ **B** 180 nmol mg min
- ☐ **C** 3 nmol mg⁻¹ min⁻¹
- ☐ **D** 180 nmol mg⁻¹ min⁻¹

(ii) Explain the results of this investigation.

Use the information in the table and your own knowledge of the Krebs cycle to support your answer.

(4)

[illegible]

(c) Fast twitch and slow twitch muscle fibres contain different numbers of mitochondria.

(i) Explain the difference in the number of mitochondria in these two types of muscle fibre.

(3)

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(ii) Describe the role of ATP in muscle contraction.

(4)

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(Total for Question 7 = 14 marks)